

Hisham Alkaed

Data Scientist | AI & NLP Engineer

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SUMMARY

Data Scientist and AI Engineer with 3+ years of experience building intelligent systems across NLP, machine learning, and knowledge graphs. Proven track record delivering production-level AI solutions within Dutch government ministries — from end-to-end RAG applications and document processing pipelines to generative AI assistants serving 100,000+ users. Multilingual (English, Dutch, Arabic, Spanish) with strong skills in translating complex technical challenges into scalable, impactful solutions.

PROFESSIONAL EXPERIENCE

Digital Business Innovator / AI Engineer Feb 2026 – Present

DMFCO & Munt Hypotheken – Part-Time

- Driving the AI transition by designing and implementing intelligent automation solutions
- Building an end-to-end Retrieval-Augmented Generation (RAG) application integrating 20+ heterogeneous data sources

Data Scientist Jan 2026 – Present

Ministerie van Justitie en Veiligheid – Datalab | Rijks-I Traineeship

- Assisting in multiple Data Science and NLP projects within the Datalab team
- Improving performance and accuracy of RAG applications through prompt engineering and retrieval optimization

AI Advisor May 2025 – Dec 2025

Ministerie van Binnenlandse Zaken en Koninkrijksrelaties – SSC-ICT | Rijks-I Traineeship

- Played a crucial role in the delivery of a generative AI assistant for 100,000+ civil servants
- Co-managed functional and technical implementation, advised stakeholders, and drove feature development
- Coordinated with internal clients and proactively translated user needs into AI-driven solutions

Data Scientist Sep 2024 – Apr 2025

Ministerie van Onderwijs, Cultuur en Wetenschap – DUO | Rijks-I Traineeship

- Built an end-to-end pipeline for scraping, parsing, anonymizing, and indexing policy documents using Python, Elasticsearch, and Streamlit
- Deployed interactive dashboards enabling instant search across 10,000+ documents
- Cut policy data retrieval time by 80%, significantly improving internal data accessibility

Linked Data Consultant Jun 2024 – Aug 2024

Taxonic

- Designed and implemented linked data solutions to enhance data interoperability and accessibility for enterprise clients
- Built knowledge graphs using semantic web technologies (SPARQL, RDF) for diverse industries including healthcare

Teaching Assistant

Sep 2022 – Jul 2023

Vrije Universiteit Amsterdam – Part-Time

- Assisted in courses: Introduction to Data Science, Databases, Intelligent Systems, and Computational Intelligence
- Graded assignments and mentored 50+ students on Python, SQL, and machine learning

Operational Management Assistant

Aug 2021 – Nov 2021

Nationale Nederlanden – Part-Time

- Automated data workflows using Power BI, reducing manual reporting time by 30%
- Built an internal room reservation system

EDUCATION

Master of Arts in Linguistics – Text Mining

2022 – 2023

Vrije Universiteit Amsterdam – GPA: 8.0/10

- Specialized in NLP, information extraction, computational linguistics, and Transformers
- Thesis: "From Words to Images: Exploring the Crucial Role of Textual Modality in Multimodal Hate Speech Detection"

Bachelor of Science in Artificial Intelligence

2019 – 2022

Vrije Universiteit Amsterdam – GPA: 8.8/10

- Core studies in machine learning, knowledge representation, cognitive computing, and data structures
- Thesis: "Entity Prediction Task in Noisy Environment"

TECHNICAL SKILLS

Languages: Python, SQL, SPARQL, R, MATLAB

AI / ML: NLP, Large Language Models, Deep Learning, Computer Vision, Transformers, RAG

Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face, spaCy, NLTK, Gensim

Data & Tools: Pandas, Elasticsearch, Streamlit, Docker, Git, Power BI, Neo4j, GraphDB

Knowledge: Knowledge Graphs, RDF, Linked Data, Ontology Engineering, Semantic Web, SPARQL

Spoken: English (fluent), Dutch (fluent), Arabic (native), Spanish (basic)

CERTIFICATIONS

Knowledge Graphs Essentials

2024

Taxonic Academy

Knowledge Graphs Foundations

2024

Taxonic Academy

SELECTED PROJECTS

Multimodal Hate Speech Detection — Python, NLP, LLMs, Fine-Tuning, ResNet, Late Fusion

Engineered a multimodal detection system for memes, fusing text and visual embeddings via late fusion with ensemble modeling.

PDFs to Insights: Information Extraction Pipeline — Python, NLP, Elasticsearch, Streamlit

End-to-end pipeline for web crawling, PDF parsing, data anonymization, and search indexing. Deployed dashboards for instant document retrieval.

Entity Prediction with Knowledge Graphs — SPARQL, Python, GraphDB, RDF

Research project on entity prediction under open-world assumptions using the Yago knowledge graph and a 20-questions game framework.

Book Recommender Application — SPARQL, GraphDB, SQL, React

Modular book recommender system connectable to any knowledge graph, built with semantic web technologies and a React frontend.